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# Persistent photoconductivity in a-IGZO thin films induced by trapped electrons and metastable donors

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Chen Wang, Wenmo Lu, Fengnan Li, Haiyue Ning, and Fei Ma[a)]

## AFFILIATIONS

State Key Laboratory for Mechanical Behavior of Materials, Xi'an Jiaotong University, No.28, Xianning West Road, Xi'an, Shaanxi 710049, People's Republic of China

[a)]**Author to whom correspondence should be addressed:** mafei@mail.xjtu.edu.cn

## ABSTRACT

Amorphous In–Ga–Zn–O (a-IGZO) thin films are prepared by pulsed laser deposition and fabricated into thin film transistors (TFTs). The concentration of oxygen vacancies in a-IGZO thin films is determined by the deposition oxygen pressure, as characterized by *in situ* x-ray photoelectron spectroscopy. The oxygen vacancies could induce persistent photoconductivity (PPC) and thus the negative shift of threshold voltage of the TFTs under illumination. The PPC in a-IGZO is quantitatively described by the time constant ($\tau$) of decay current. The continuous illumination could cause a fast decay ($\tau \sim 0.1$ s) and a slow decay ($\tau \sim 100$ s); however, the pulsed laser (20 ns duration) just results in the fast decay ($\tau \sim 0.1$ s). The fast decay is temperature independent and can be ascribed to the transition of trapped electrons at 0.035–0.75 eV below the conduction band. The slow decay occurs at 210 K or above, resulting from the generation of metastable donors at 0.9 and 19.3 meV below the conduction band. The thermal activation energies required for the generation and annihilation of the metastable donors are 2.2 and 375 meV, respectively. The spectrum response of photocurrent (600–300 nm) and density functional theory calculation illustrate that the metastable donors are transformed from the neutral oxygen vacancies at a highly localized valence band tail.

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## I. INTRODUCTION

As the channel material candidate in thin film transistors (TFTs), amorphous In–Ga–Zn–O (a-IGZO) exhibits high field-effect mobility, good subthreshold swing, low temperature processing capability, large on–off ratio, and transparency to visible light, and it has been extensively studied for the next-generation liquid crystal displays (LCDs) and organic light-emitting diode displays (OLEDs).[1] Although high performance a-IGZO TFTs and the carrier transport mechanism have been explored,[2–6] the nature of the defects in a-IGZO and their influences on the light response are not well understood;[7–9] for instance, the threshold voltage ($V_{th}$) will shift negatively under illumination and it will take a long time for $V_{th}$ recovery. Moreover, this performance degradation will be further exacerbated by negative gate bias stress,[10,11] which is adverse to the application of the a-IGZO TFT based flat-panel displays and needs to be completely explored.

The instability of a-IGZO TFTs under illumination might be related to persistent photoconductivity (PPC).[12,13] The PPC has been observed in III–V and II–IV compound semiconductors and normally attributed to two mechanisms.[14–16] First, free carriers can transit between the trap states and the conduction band. As a result, the direct band-to-band recombination is suppressed and the lifetime of free carriers is extended, which is evidenced in GaN,[17] $TiO_2$,[18] and SiC.[19] Second, PPC can be induced by the transformation of defects from deep to shallow energy states. For example, Si dopants in AlGaAs could create negatively charged deep states,[20] which could be excited into shallow donors and provide extra free carriers. After the light is turned off, the energy barrier of ~0.2 eV prevents the transition from shallow donors back to deep states. Yasuno *et al.*[21] excited a-IGZO sample with a short laser pulse (100 ns) by a time-resolved microwave photoconductivity decay technique (μ-PCD). Only a fast decay of about 4 $\mu$s was observed, and they ascribed it to the transition of trapped electrons. Lee *et al.*[12,13] observed an extremely slow photocurrent decay (~1000 s) in the steady-state measurement and attributed it to the metastable states of weakly bonded oxygen. Hossain Chowdhury *et al.*[22] carried


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out a temperature dependent experiment and demonstrated that PPC was caused by the generation and annihilation of a double donor related to oxygen vacancies. Jang *et al.*[23] ascribed the PPC to the shallow donors of peroxides. Apparently, the origins of PPC in a-IGZO thin films are still unclear.

In this study, a-IGZO thin films are prepared by pulsed laser deposition (PLD) and fabricated into TFTs. The photoelectric properties of a-IGZO thin films are studied by illumination with continuous light and 248 nm pulsed laser. Temperature varying experiments at 13–310 K, spectrum response, and density functional theory (DFT) calculations are performed to analyze the transition of electrons and the generation of defect states in a-IGZO thin films. Accordingly, the mechanism of PPC is discussed.

## II. EXPERIMENT

### A. Preparation and characterization of a-IGZO thin films

40 nm a-IGZO thin films are prepared by PLD with a KrF excimer laser (248 nm, ~10 J $cm^{-2}$ per pulse). The pulse number is 500 and the repetition frequency is 2 Hz. During pulsed laser deposition, the substrates are attached to a temperature control platform, which is cooled by 20 °C circulating water and monitored by Shimaden SRS10 PID through a K-type thermocouple. Therefore, the temperature of the substrates can be maintained at 20 ± 0.5 °C. A homemade crystalline IGZO ceramic (In:Ga:Zn:O = 2:2:1:7 in atomic ratio) is used as the target. The PLD chamber is inlet with 2.7–3.5 Pa pure oxygen (99.999%). During deposition, the substrates are mounted on a disk rotating at a fixed speed, and the substrates are arranged on a ring, as illustrated in Fig. S1 in the supplementary material, so that the plasma ablated from the target is uniformly distributed in the deposition space. The surface morphology of the a-IGZO thin films is observed by a scanning electron microscope (SEM, FEI-VERIOS 460, with an acceleration voltage of kV and a probe current of 25 pA), and the phase structure is characterized by x-ray diffraction (XRD, D8 advance, λ = 1.5406 Å). An interconnected x-ray photoelectron spectroscopy (XPS) and PLD system in a vacuum chamber is used for *in situ* characterization of elemental contents and chemical states to avoid the contaminant and adsorbate, and the excitation source is Al Kα with an energy of 1486.6 eV.

### B. Fabrication of TFTs and electrical measurement

Figure S2 in the supplementary material illustrates the device architecture, which is a typical top contact and bottom gated structure. The a-IGZO channel layer is deposited on the $p^+$-Si substrate with 280 nm thermally oxidized $SiO_2$ by PLD and patterned by stainless steel shadow mask. The $p^+$-Si substrate and 280 nm thermally oxidized $SiO_2$ act as the gate electrode and the gate insulator, respectively. Then, the source/drain electrodes made of 100 nm Mo thin films are deposited by RF magnetron sputtering and patterned by a stainless steel shadow mask. The channel width and length of the devices are 2000 and 300 $\mu$m, respectively. The patterned a-IGZOs just serve as the channels, so the lateral fringing current is suppressed. In our experiment, eight groups of a-IGZO TFTs are fabricated for a given deposition oxygen pressure in the same batch and three batches are repeated for the same set of parameters, so at least 120 groups of TFTs are tested in this study. Transfer curves of each TFT are measured in dark to evaluate the device consistency, and at least three groups of TFTs in the same batches are tested to study the instability under light illumination. The temperature varying experiment is carried out with four TFTs deposited at 3.1 Pa oxygen pressure. Figure S3 in the supplementary material displays the transfer curves of eight groups of TFTs from the same batch, and six of them possess almost the coincided transfer curves, proving good replicability and reliability. Transfer curves are measured by sweeping $V_g$ from −60 to 60 V in steps of 0.5 V, with $V_d$ kept at 2 V. All the electrical tests are carried out in a vacuum chamber (~$10^{-3}$ Pa by turbo pump). The illumination is introduced by an optical window on the vacuum chamber.

### C. Photocurrent measurement

The photocurrent is measured by exposing the a-IGZO TFTs to continuous monochromatic light (600–300 nm) and pulsed laser (248 nm) with the gate electrode floating. The continuous monochromatic light is generated by a mercury–xenon lamp with a monochromator. The pulsed laser with duration of 20 ns is provided by the KrF laser with the energy intensity of 10–30 mJ $cm^{-2}$. The illumination intensity is calibrated by a Thorlabs PM100D power and energy meter. The temperature dependent electrical properties are measured in a vacuum chamber at 13–310 K using a Helium compressor (APD HC-2). The sample is illuminated through the quartz window. The measurement voltage is 2 V, and the current sampling with an interval of 0.02 and $10^{-5}$ s is acquired by a Keithley 2612B source meter and 6500 DMM ADC, respectively.

### D. DFT calculation

The a-IGZO model containing 192 atoms is generated by the melt-and-quench process in *ab initio* molecular dynamics (MD) simulation, in which the GGA function is used for the exchange–correlation potential and the projector-augmented wave potential in the VASP code. An energy cutoff is set at 400 eV, and a k-point set by the 3 × 3 × 3 Monkhorst–Pack mesh is used for Brillouin-zone integration. The force convergence condition of 0.05 eV $Å^{-1}$ is used. The crystalline $In_2Ga_2ZnO_7$ supercell is melted at 3000 K for 20 ps, then quenched rapidly down to 300 K at a rate of 250 K $ps^{-1}$, and relaxed at 300 K for 10 ps, with a time step of 2 fs. A canonical ensemble is used by a Nose thermostat. The GGA + U method is used and the U values of 7.0, 8.0, and 8.0 eV are set for the In4d, Ga3d, and Zn3d orbitals, respectively.[24,25] Figure S4(a) in the supplementary material shows the model of a-IGZO, and Fig. S4(b) in the supplementary material shows the pair-correlations function (PCF) of In, Ga, and Zn. The bond lengths of In–O, Ga–O, and Zn–O are 2.17, 1.92, and 1.98 Å, respectively, which is consistent with the values evaluated by EXAFS measurements.[4] The localization of an orbital $\psi_n(\mathbf{r}_i)$ is described by the inverse participation ratio (IPR), $I(\psi_n)$ as

$$I(\psi_n) = N \frac{\sum_{i=1}^{N} |\psi_n(\mathbf{r}_i)|^4}{\left(\sum_{i=1}^{N} |\psi_n(\mathbf{r}_i)|^2\right)^2}, \quad (1)$$

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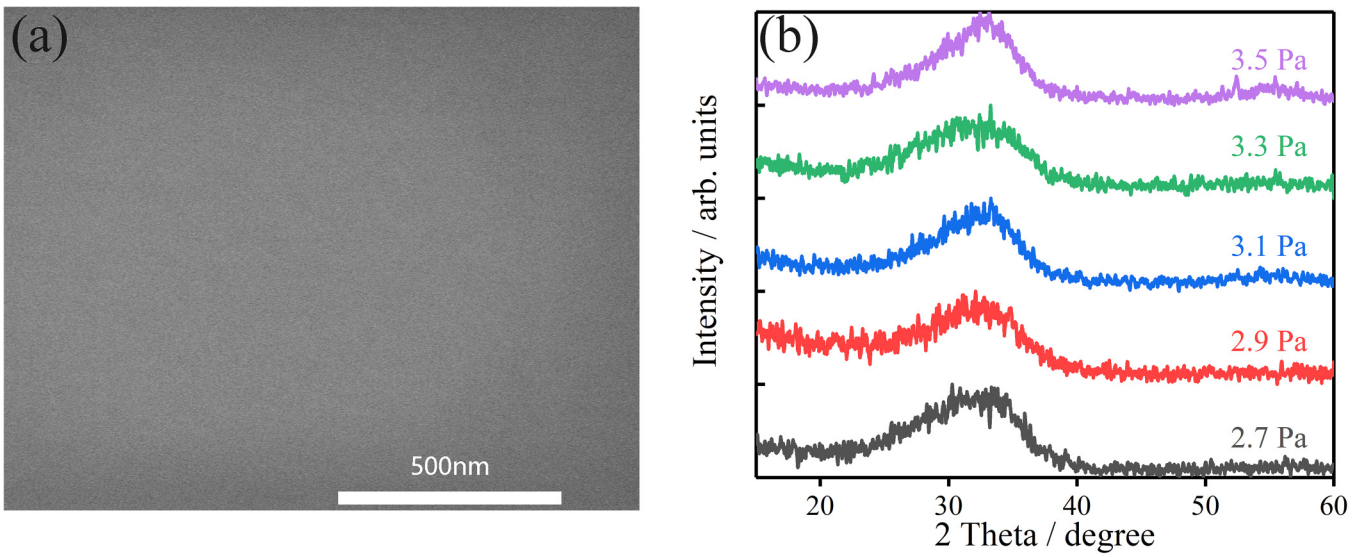



**FIG. 1.** (a) The SEM image of the a-IGZO thin film deposited at 3.1 Pa oxygen pressure. (b) XRD patterns of a-IGZO thin films deposited at different oxygen pressures.

in which N is the number of atoms in the cell and i is the index of the atom. The large IPR corresponds to highly localized states (<N), and the value 1 presents the totally delocalized states.[26]

## III. RESULTS AND DISCUSSION

### A. Characterization of the a-IGZO thin films

The surfaces of the films deposited at 2.7–3.5 Pa oxygen pressure are all smooth, indicating the high quality of the a-IGZO thin films. Figure 1(a) shows the typical SEM image of the a-IGZO thin film deposited at 3.1 Pa oxygen pressure. Figure 1(b) shows the XRD patterns of a-IGZO thin films deposited at 2.7–3.5 Pa oxygen pressure. No sharp peak can be identified in all the patterns, suggesting deposition pressure independent amorphous feature of a-IGZO thin films. The atomic ratio of In:Ga:Zn in a-IGZO is determined to be 2.12 ± 0.20:2.16 ± 0.17:1 by *in situ* XPS, which is very close to the atomic percentage in the target (2:2:1). The specific atomic percentages are listed in Table SI in the supplementary material. Figure 2(a) displays the XPS spectra of In $3d^5/_2$ in the a-IGZO films deposited at 2.7–3.5 Pa oxygen pressure, and the In $3d^5/_2$ spectrum can be deconvoluted into two peaks at 444.7 eV ($In_I$ peak) and 443.4 eV ($In_{II}$ peak) by Voigt function [Fig. 2(b)]. The FWHM of In $3d^5/_2$ in a-IGZO is 1.6 eV and the two sub-peaks of In $3d^5/_2$ are 1.3 eV apart, which is large than the half of the FWHM of In $3d^5/_2$ in a-IGZO. According to the Rayleigh criterion, the two sub-peaks can be distinguished. The $In_I$ and $In_{II}$ peaks are ascribed to $In^{3+}$ (indium oxide) and metallic In, respectively. The metallic In was evidenced in pure $In_2O_3$ and ITO.[27–31] The existence of metallic In is also confirmed by *ab initio* molecular dynamics (MD) simulation in VASP. Figure S5 in the supplementary material displays the pair-correlations function (PCF) of In–In in a-IGZO, crystalline IGZO and metallic In. The nearest distance between In atoms in a-IGZO is 2.72 Å, which is smaller than that in crystalline IGZO (3.44 Å) and metallic In (3.26 Å). That is, some In 5s orbits in a-IGZO are strongly overlapped, resulting in metal-like chemical states. The existence of metallic In illustrates the formation of oxygen vacancies.[31] As shown in the inset of Fig. 2(b), the concentration of oxygen vacancies, that is, the $In_{II}/(In_I + In_{II})$ ratio, decreases with increasing deposition oxygen pressure. Consequently, the oxygen vacancies can be compensated by high deposition oxygen pressure to some degree.

### B. TFTs instability under illumination

The a-IGZO TFTs deposited at 2.7–3.5 Pa oxygen pressure are illuminated by a continuous light with 300 nm wavelength and 1 mW $cm^{-2}$ energy intensity for 10 s. Figure 3 shows the transfer curves of the a-IGZO TFTs before illumination, upon illumination, as well as 100 s later after illumination. The threshold voltage ($V_{th}$) and the flatband current ($I_{flat}$) are extracted from the transfer curves. The transfer curves are negatively shifted under illumination and could be restored in dark. Upon illumination, the $V_{th}$ is negatively shifted by −7.1, −5.2, −3.5, −2.1, and −1.4 V for the IGZO thin films deposited under 2.7, 2.9, 3.1, 3.3, and 3.5 Pa oxygen pressure, respectively. Table SII in the supplementary material summarizes the mobilities, subthreshold swings, on/off ratios, and turn-on voltages before illumination, upon illumination, and 100 s after illumination. The lower the deposition oxygen pressure is, the larger the negative $V_{th}$ shift is. However, the subthreshold performance of the transfer curves is well maintained, suggesting that illumination does not induce defect states around the Fermi level. Therefore, the $V_{th}$ shift can be attributed to the extra photo-generated free carriers. From the transfer curves of the IGZO TFTs 100 s later after illumination, it is found that the recovery of the negative $V_{th}$ shift is a rather slow process, implying the long lifetime of

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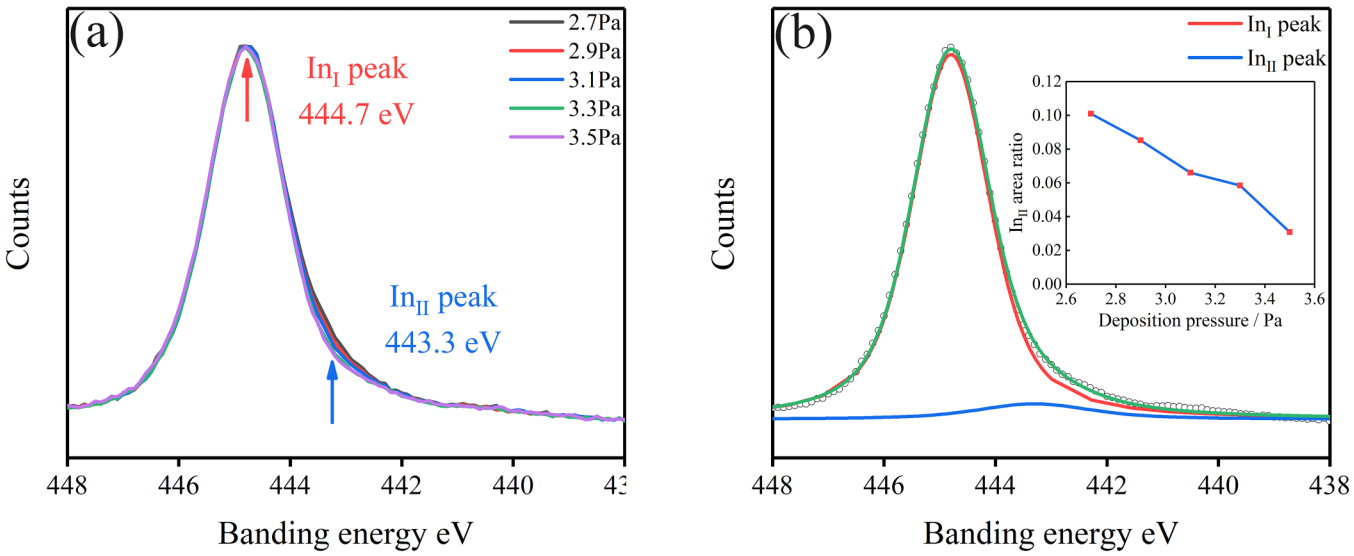



**FIG. 2.** The XPS analysis of a-IGZO thin films. (a) In $3d^5/_2$ spectra of a-IGZO thin films deposited at 2.7–3.5 Pa oxygen pressure. (b) In $3d^5/_2$ spectrum of a-IGZO thin films deposited at 3.1 Pa oxygen pressure. The inset is the area ratio of $In_{II}$ peak with deposition pressure.

photo-generated free carriers. Accordingly, the PPC might lead to the instability of TFTs under illumination and will be further studied by photocurrent measurement.

### C. Photocurrent induced by continuous light

Figure 4(a) shows the photocurrent in the a-IGZO thin films deposited at 2.7–3.5 Pa oxygen pressure, illuminated by a continuous light with 300 nm wavelength and 1 mW cm$^{-2}$ energy intensity for 10 s. After the light is turned off, the photocurrent exhibits a multi-exponential decay. The photocurrent is larger in the a-IGZO thin films with more oxygen vacancies, proving that PPC originates from the oxygen vacancies. The decay current can be well fitted by the following equation:

$$I_{decay} = I_0 + A_1 \exp\left(-\frac{t}{\tau_1}\right) + A_2 \exp\left(-\frac{t}{\tau_2}\right), \quad (2)$$

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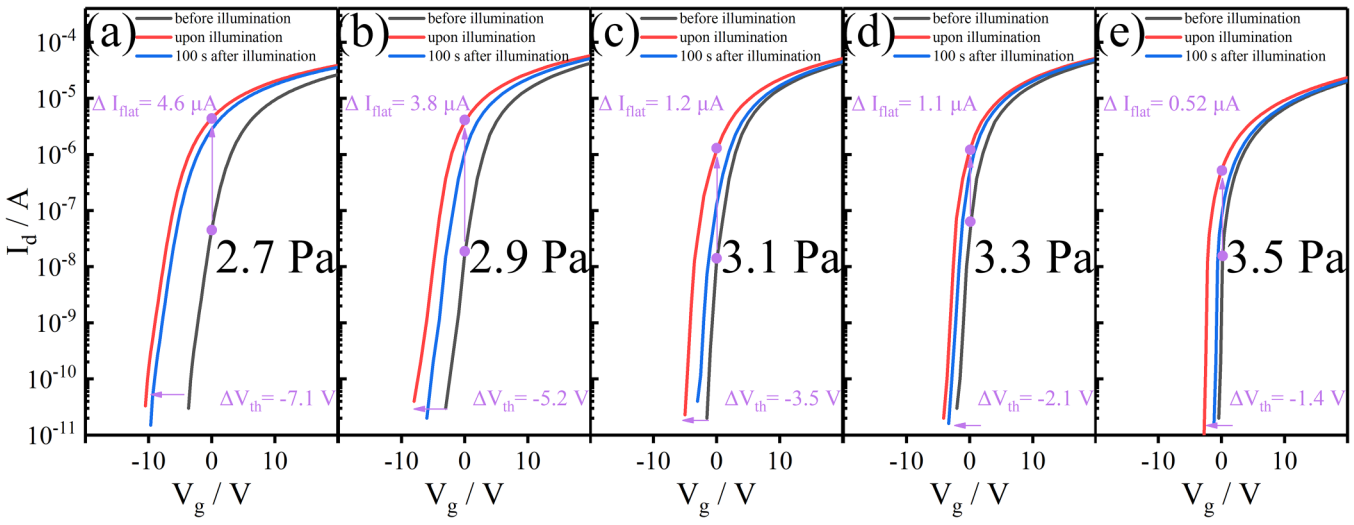



**FIG. 3.** The transfer curves of a-IGZO TFTs before illumination, upon illumination, as well as 100 s after illumination. The a-IGZO thin films are deposited at different oxygen pressures: (a) 2.7, (b) 2.9, (c) 3.1, (d) 3.3, and (e) 3.5 Pa, illuminated by a continuous light with 300 nm wavelength and 1 mW cm$^{-2}$ energy intensity for 10 s.


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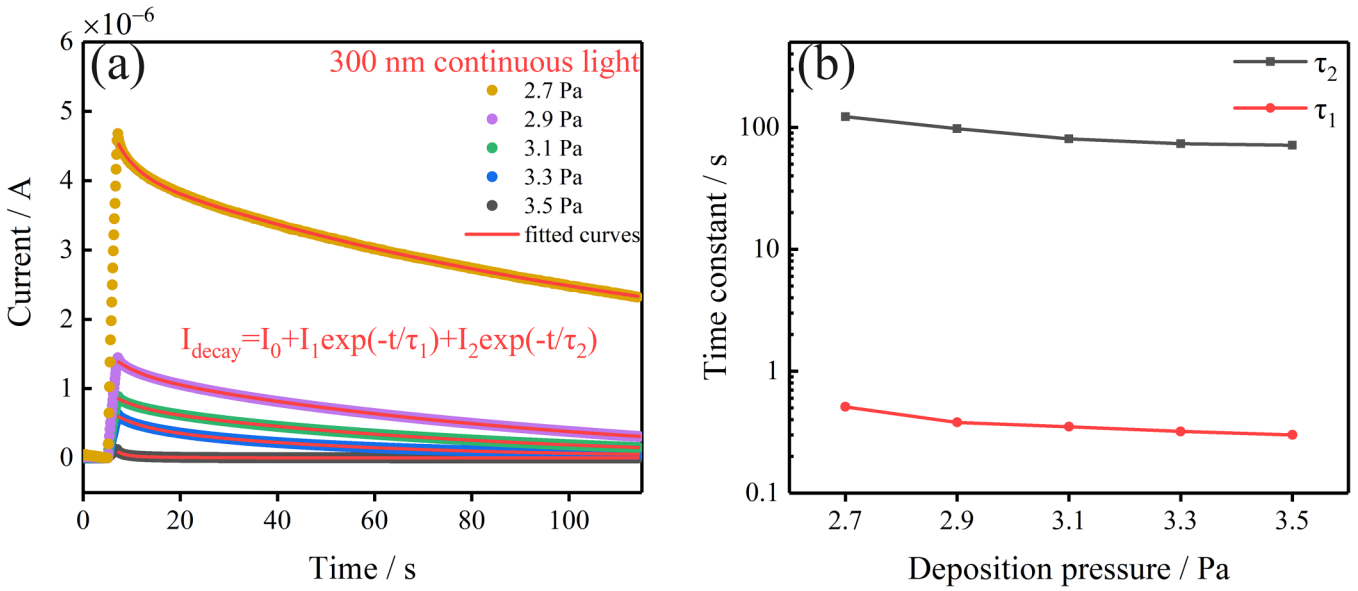



**FIG. 4.** (a) The photocurrent of a-IGZO thin films deposited at 2.7–3.5 Pa oxygen pressure, upon illumination by a continuous light with 300 nm wavelength and 1 mW $cm^{-2}$ intensity for 10 s. (b) The extracted time constants of decay current.

in which $I_0$ is the dark current, t is the time, $A_1$ and $A_2$ are the amplitudes, and $\tau_1$ and $\tau_2$ are the time constants. As shown in Fig. 4(b), a fast decay with $\tau_1 \sim 0.1$ s and a slow decay with $\tau_2 \sim 100$ s can be extracted for each sample. The time constants $\tau_1$ and $\tau_2$ are nearly unchanged with the deposition pressure, implying that the decaying mechanism should not be affected by the concentration of oxygen vacancies.

### D. Photocurrent induced by pulsed laser

Figure 5(a) plots the photocurrent of a-IGZO thin films deposited at 3.1 Pa oxygen pressure, induced by 248 nm pulsed laser with the energy intensities of 10, 20, and 30 mJ $cm^{-2}$. As compared with that illuminated by continuous light, the pulsed laser induced photocurrent in a-IGZO thin films decays more rapidly down to the initial current in dark. As shown in Fig. 5(b), the transient photoelectrical current can be fitted by four exponential functions and can be divided into two stages: a direct combination[21] ($\tau_1 \sim 3.0 \times 10^{-4}$ and $\tau_1 \sim 3.0 \times 10^{-3}$ s) and a decay process ($\tau_3 \sim 0.45$ and $\tau_4 \sim 3.54$ s). Because of the short duration and high energy intensity of pulsed laser, abundant free carriers are excited onto the conduction band. After the laser pulse, most of the free carriers will be annihilated through the direct band-to-band recombination.[21] The decay described by $\tau_3$ and $\tau_4$ is related to PPC, which can also be observed in the photocurrent illuminated by the continuous light, but the slow decay with $\tau \sim 100$ s cannot be induced by the pulsed laser. Therefore, the fast decay is irrelevant to the illumination duration and might be caused by the intrinsic defects in a-IGZO. However, sufficient illumination time is required for the slow decay; thus, the slow decay might result from the defects generated during illumination.

### E. Photocurrent in temperature varying experiments

Temperature varying experiments are carried out at 13–310 K to understand the mechanism of PPC. The representative a-IGZO thin films deposited at 3.1 Pa oxygen pressure are chosen for temperature varying measurements. Figure 6 plots the photocurrent of the a-IGZO TFTs under illumination by continuous light with 300 nm wavelength and 1 mW $cm^{-2}$ energy density for 10 s. The multi-exponential time constant extraction method[32] is adopted to analyze the decay current, and the time constants are evenly sampled from $10^{-2}$ to $10^{6}$ s in the logarithmic scale,

$$I_{decay} = I_0 + \sum_i A_i \exp\left(-\frac{t}{\tau_i}\right), \quad (3)$$

in which $\tau_i$ is the time constant, $A_i$ is the amplitude, and $I_0$ is the dark current. Figure 7 shows the time constant spectra. The fast decay with $\tau \sim 0.1$ s exists at all temperatures but the slow decay with $\tau \sim 100$ s occurs only at the temperature higher than 150 K, that is, PPC should be controlled by two mechanisms.

### F. Transition of trapped electrons

Based on the time constant spectra, the direct band-to-band recombination with $\tau \sim 10^{-4}$ s is not observed, but PPC with $\tau \sim 0.1$ s still exists, which might be attributed to the transition of trapped electrons onto the conduction band.[21,33–35] As reported previously, a series of localized states exist at 0.1–1.0 eV below the conduction band in a-IGZO.[36] Those localized states are not the recombination centers and will not shorten the lifetime of carriers but can trap carriers and lead to PPC.[13] Figure 8(a) shows the

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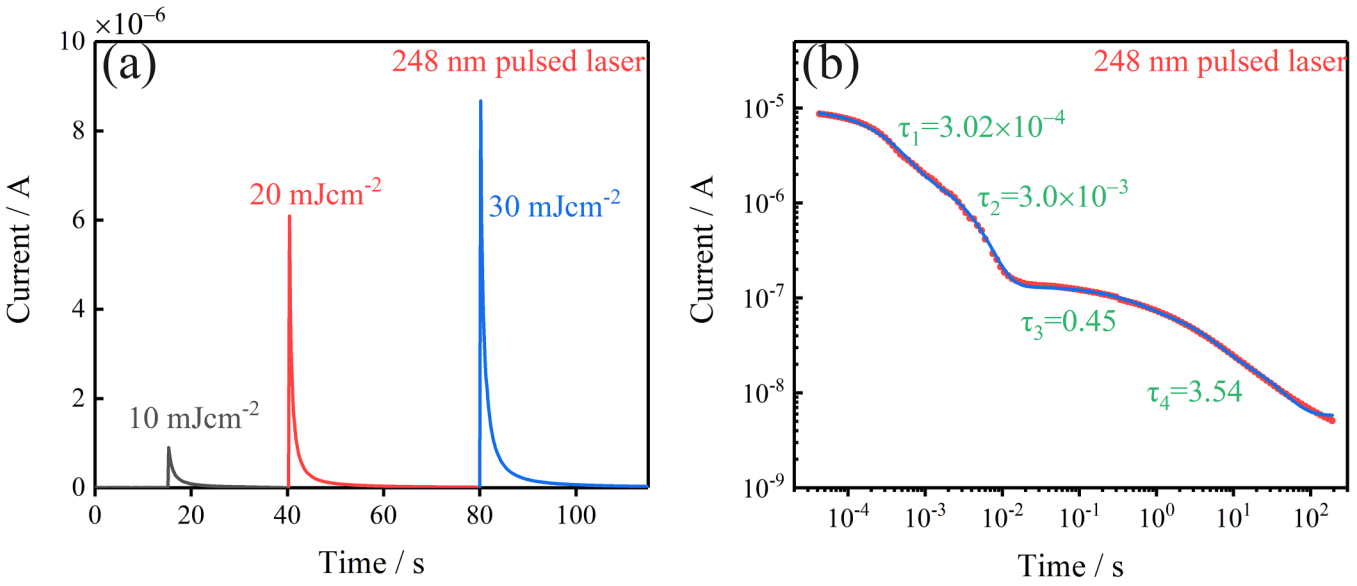



**FIG. 5.** (a) The photocurrent of the a-IGZO thin films deposited at 3.1 Pa oxygen pressure upon illumination by 248 nm pulsed laser with energy intensities of 10, 20, and 30 mJ $cm^{-2}$. (b) The transient decay current of the a-IGZO thin films deposited at 3.1 Pa oxygen pressure after illumination by a pulsed laser with energy intensity of 30 mJ $cm^{-2}$.

principal transitions in a single trap state model[37] with the generation rate of free carriers and trapped electrons described as

$$\frac{dn}{dt} = C - D - G, \tag{4}$$

$$\frac{dn_T}{dt} = -C + D, \tag{5}$$

in which n is the density of free carriers, $n_T$ is the density of trapped electrons, t is the time, C is the release rate, D is the

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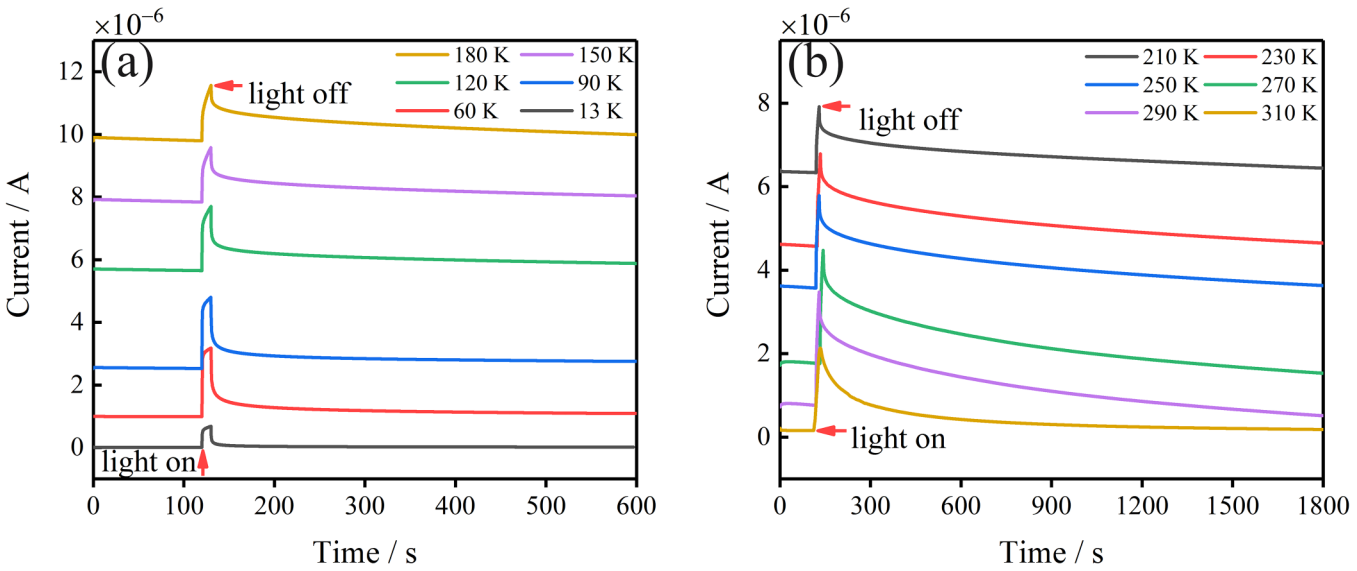



**FIG. 6.** The photocurrent of the a-IGZO thin films deposited at 3.1 Pa oxygen pressure, measured at (a) 13–180 K and (b) 210–310 K, upon illumination by a continuous light with 300 nm wavelength and 1 mW $cm^{-2}$ intensity for 10 s.


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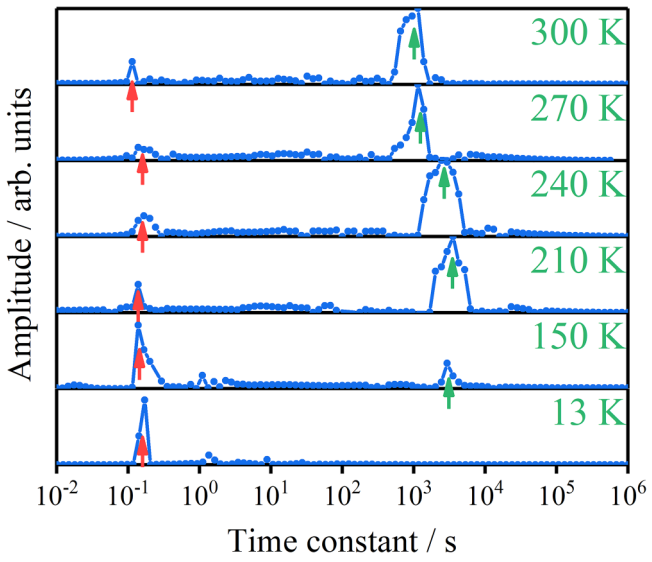



**FIG. 7.** The time constant spectra from $10^{-2}$ to $10^{6}$ s in the logarithmic scale by the multi-exponential time constant extraction method.

trapping rate, and G is the direct recombination rate. The density of free carriers, n, can be calculated by $\sigma = ne\mu$, in which $\sigma$ is the conductivity, e is the electron charge and $\mu$ is the mobility ($\sim 10\ cm^2\ V^{-1}\ s^{-1}$). The release, trapping, and direct recombination rates can be written as[38]

$$C = n_T v \exp\left(-\frac{E_i}{k_B T}\right), \tag{6}$$

$$D = n\sigma u N_T, \tag{7}$$

$$G = Rn, \tag{8}$$

in which v is the attempt-to-escape frequency ($10^{13}\ s^{-1}$, close to the maximum phonon frequency of ZnO[13,25,34,39]), $E_i$ is the energy level of the trap state from conduction band, $\sigma$ is the capture cross section ($9.0 \times 10^{-20}\ cm^{-2}$, consistent with the experimental data[35,40]), u is the electron thermal velocity $\sqrt{\frac{8k_BT}{\pi m_n^*}}$, $m_n^*$ is the effective carrier mass ($0.34\ m_e$),[41] $N_T$ is the density of trap states, and R is the direct recombination rate. Thus, the energy level $E_i$ and the density of the trap states $N_T$ can be acquired through numerical simulations. Figure 8(b) plots the fitted free carrier density and trapped electron density at 13 K. The initial current drop is due to the electrons captured at the trap states and the subsequent slow decay is owing to the release of trapped electrons onto the conduction band. Since the release rate (D) is strongly dependent on the temperature, only the carriers trapped in shallow states can be excited onto the conduction band at low temperature. As the temperature is elevated, the electrons trapped at the deep states can be activated. Figure 8(c) displays the trap states fitted from the photocurrent at 13–310 K. They are in the energy range of 0.035–0.75 eV below the conduction band and have the highest density of states at 0.1 eV, similar to the value measured by μ-PCD.[21] The Laplace transform method[13,18,42] is also adopted to extract the density of trap states. The free carrier density can be described by the integration of a continuous trap distribution,

$$n = n_0 + \int g(E) \exp\left(-\frac{t}{\tau(E)}\right) dE, \tag{9}$$

in which $n_0$ is the dark carrier density, $\tau(E) = \frac{1}{v} \exp\left(\frac{E}{kT}\right)$ is the time constant, g(E) is the density of traps, and E is the energy level below the conduction band. The decay current can be expanded as

$$n = n_0 + \sum N_i \exp\left(-\frac{t}{\tau_i}\right), \tag{10}$$

and the density of trap states can be evaluated by Laplace transformation as

$$g(E) = \sum \frac{N_i}{k_B T} \frac{v \exp\left(-\frac{E}{k_B T}\right)}{\tau_i \left(v \exp\left(-\frac{E}{k_B T}\right) + \frac{1}{\tau_i}\right)^2}. \tag{11}$$

Twenty exponential functions are used to reproduce the decay current at 13–310 K, and the extracted density of states is plotted in Fig. 8(d). The density of trap states extracted by Laplace transformation is consistent with that obtained by numerical simulation. Accordingly, the PPC with $\tau \sim 0.1$ s could be well understood by the transition of trapped electrons.

### G. Generation and annihilation of metastable donor states

As shown in Fig. 6, after the illumination by continuous light, a slow decay is observed at 210 K or higher. Since long illumination duration is needed to induce the slow decay, it might be due to the metastable donors generated upon illumination.[22,23,43] As shown in Fig. 9(a), when the a-IGZO thin films are illuminated, the photocurrent increases sharply in 0.02 s or shorter. After that, the current increases linearly with time, and the higher the temperature is, the more rapidly the current increases. After the light is turned off, the photocurrent decreases sharply but with a residual current, as shown in Fig. 9(b). The residual current almost is equal to the increase of photocurrent in the linear region, and can be ascribed to the metastable donors generated upon illumination, but the density of metastable donors cannot be changed at 90 K or below.

For a given density of donors, free carriers increase with the temperature as a result of thermal activation, so the residual current is determined by the density of metastable donors as well as the temperature. The donor level could be evaluated from the thermally activated current.[44] First, the IGZO thin films are illuminated by a 300 nm continuous light at 13 K for 30 s to induce metastable donor states. Second, the light is turned off to approach a

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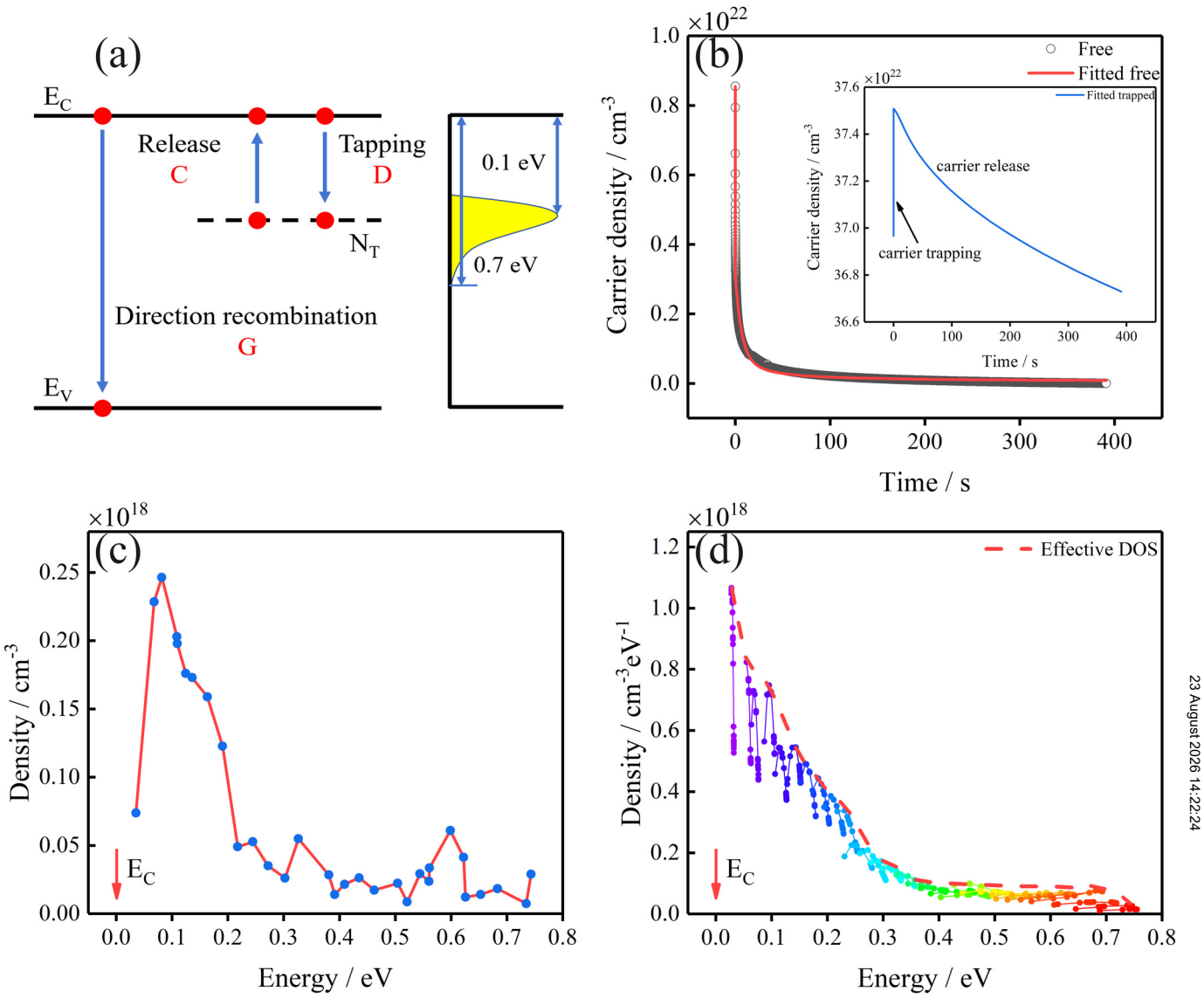



**FIG. 8.** (a) Schematic diagram of a trap states in the bandgap of a semiconductor. (b) The free carrier density of the a-IGZO thin films deposited at 3.1 Pa oxygen pressure, measured at 13 K, upon illumination by a continuous light with 300 nm wavelength and 1 mW cm$^{-2}$ intensity for 10 s. The inset is the fitted trapped electron density. The trap state density extracted by (c) the numerical simulation, and (d) the Laplace transform method.

constant dark current. Then, the temperature of the IGZO thin films is gradually elevated up to 90 K at a rate of 10 K min$^{-1}$, and the dark current is measured at each temperature. The probability of electron occupying the donor level is described by Fermi–Dirac distribution,

$$f_D = \frac{1}{1 + \exp\left(\frac{E_D - E_F}{kT}\right)}, \tag{12}$$

in which $E_D$ is the donor level, $E_F$ is the Fermi level, k is the Boltzmann constant, and T is the temperature. The density of free carriers in conduction band is

$$n = N_D - N_D f_d = \frac{N_D}{1 + \exp\left(-\frac{E_D - E_F}{kT}\right)}, \tag{13}$$

where $N_D$ is the density of donor. At the low temperature, the

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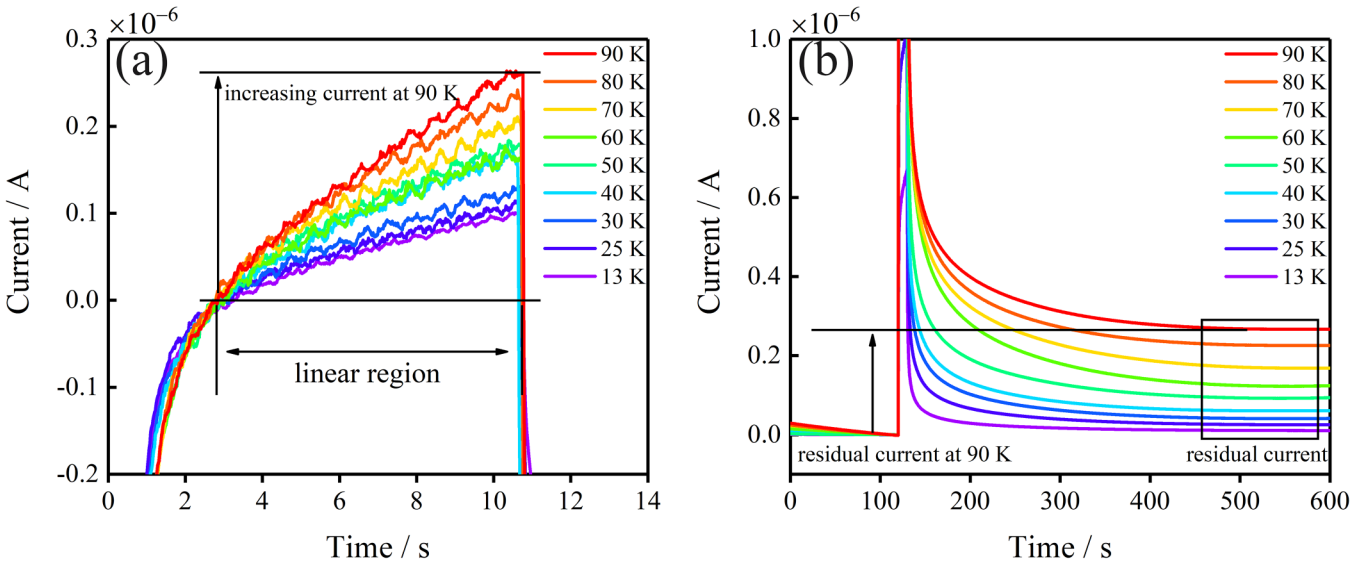



**FIG. 9.** The photocurrent (a) generation and (b) decay of the a-IGZO thin films deposited at 3.1 Pa oxygen pressure, measured at 13–90 K, upon illumination by a continuous light with 300 nm wavelength and 1 mW $cm^{-2}$ intensity for 10 s. The initial dark current of the decay processes is scaled to zero for comparison.

carriers in conduction band are totally from the donor level, which can be written as

$$n = N_c \exp\left(-\frac{E_c - E_F}{kT}\right), \tag{14}$$

where $N_c$ is the effective density of conduction band and $E_c$ is the conduction band level. From Eqs. (13) and (14), the Fermi level and $N_c$ can be evaluated as

$$E_F = \frac{1}{2}(E_c + E_D) + \frac{1}{2}\log\left(\frac{N_D}{N_C}\right)kT, \tag{15}$$

$$N_C = 2\left(\frac{m_n^* kT}{2\pi\hbar^2}\right)^{\frac{3}{2}}, \tag{16}$$

where $\hbar$ is the reduced Planck constant. Then, the carrier density in conduction band is

$$n = (N_C N_D)^{\frac{1}{2}} \exp\left(-\frac{E_C - E_D}{kT}\right) = 2^{\frac{1}{2}}\left(\frac{m_n^* kT}{2\pi\hbar^2}\right)^{\frac{3}{4}} N_D^{\frac{1}{2}} \exp\left(-\frac{\Delta E_D}{kT}\right). \tag{17}$$

However, the dark current cannot be fitted well by a single donor level, and the model containing two donor levels is required and the carrier density is rewritten as

$$n = 2^{\frac{1}{2}}\left(\frac{m_n^* kT}{2\pi\hbar^2}\right)^{\frac{3}{4}}\left(N_{D_1}^{\frac{1}{2}} \exp\left(-\frac{\Delta E_D 1}{kT}\right) + N_{D_2}^{\frac{1}{2}} \exp\left(-\frac{\Delta E_D 2}{kT}\right)\right). \tag{18}$$

The fitted dark current is plotted in Fig. 10. Two shallow donors are extracted at 0.9 and 19.3 meV below conduction band. Supposing the constant ratio of the two donors at different temperatures, the density of metastable donors can be calculated from Fig. 9(b), and the results are plotted in Fig. 11(a), which is fitted by the Arrhenius function $N_{residual} = A\exp\left(-\frac{Ea}{k_B T}\right)$. The thermal activation energy Ea for the generation of metastable donors is fitted to be 2.2 meV.

The annihilation of metastable donors can take place at 210 K or higher and is also a thermally activated process, which can be described as

$$\frac{dN_D}{dt} = -AN_D \exp\left(-\frac{Ea}{k_B T}\right), \tag{19}$$

$$N_D = B\exp\left(-\frac{t}{\tau}\right), \tag{20}$$

in which A and B are coefficients. The time constant $\tau$ can be evaluated as

$$\tau^{-1} = A\exp\left(-\frac{Ea}{k_B T}\right). \tag{21}$$

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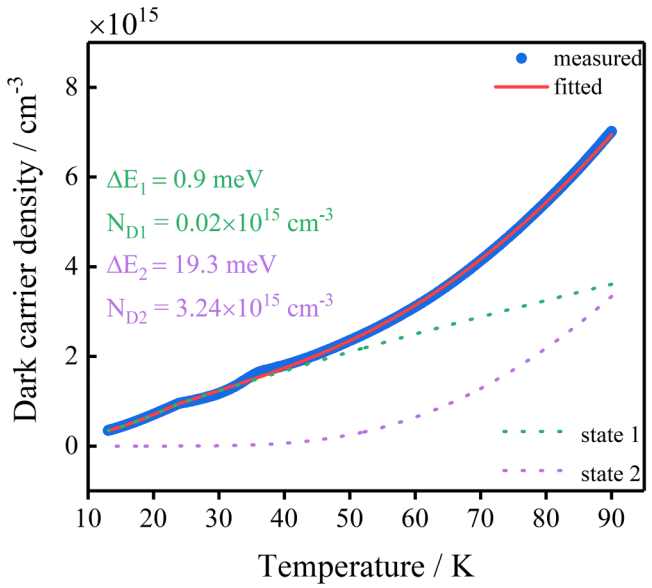



**FIG. 10.** The dark current of the a-IGZO thin films deposited at 3.1 Pa oxygen pressure, measured at 13–90 K, upon illumination by a continuous light with 300 nm wavelength and 1 mW cm$^{-2}$ intensity for 30 s, and two extracted metastable donor states.

According to Fig. 6(b), the inverse of time constant ($\tau^{-1}$) at 210–310 K is calculated, and the thermal activation energy of annihilation is fitted to be 375 meV, as shown in Fig. 11(b). A very high transformation energy suggests that the recovery from $V_O^{2+}$ to $V_O^{*2+}$ is strongly dependent on the temperature. As shown in Fig. 7, the transformation from $V_O^{2+}$ to $V_O^{*2+}$ cannot take place at 150 K or below, even under light illumination. However, the reversible transformation between $V_O^{2+}$ and $V_O^{*2+}$ is evidenced if the temperature is elevated up to 200 K or higher, as illustrated in Fig. 4(a). Such a transformation of illumination-induced metastable defects in a-IGZO has also been reported.[22,45] Consequently, long-duration PPC occurs in the a-IGZO thin films, which might usually result in instability of TFTs under illumination.

### H. Metastable donor states induced by oxygen vacancies

Figure 12(a) shows the spectrum response of photocurrent in the range of 600–300 nm (2.07–4.13 eV). As displayed in Fig. 12(b), the photocurrent can be induced by the light with photon energy smaller than the bandgap of a-IGZO (~3.1 eV), and the time constants of the photocurrent decay are almost independent of the photon energy. The photocurrent is related to the light absorption as[12]

$$\sigma_{ph} = \eta e \mu \tau F (1 - R)(1 - e^{-\alpha d})/d, \quad (22)$$

in which $\eta$ is the quantum efficiency (~1), $\mu\tau$ is the product of mobility and lifetime of the photo-generated carriers (0.2 cm$^{-2}$ V$^{-1}$),[12] R is the measured reflectance, $\alpha$ is the absorption coefficient, and d is the thickness of a-IGZO thin films. The absorption coefficient spectra are extracted from the spectrum responses [Fig. 12(c)]. The absorption coefficient at 3.1 eV (400 nm) or higher can be well fitted by the Tauc function, and the optical bandgap is fitted to be 3.06 eV. The tail-like absorption

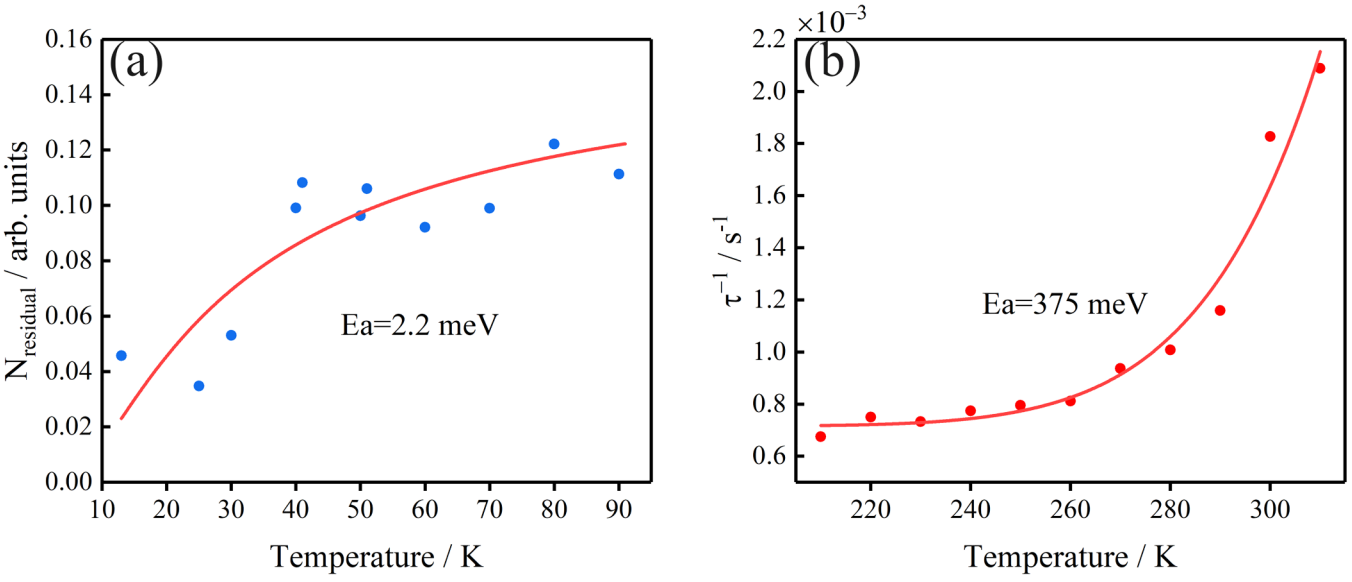



**FIG. 11.** (a) The residual density of metastable donors measured at 13–90 K and (b) the reciprocal time constant of the metastable donor recombination of the a-IGZO thin films deposited at 3.1 Pa oxygen pressure, measured at 210–310 K, upon illumination by a continuous light with 300 nm wavelength and 1 mW cm$^{-2}$ intensity for 10 s.

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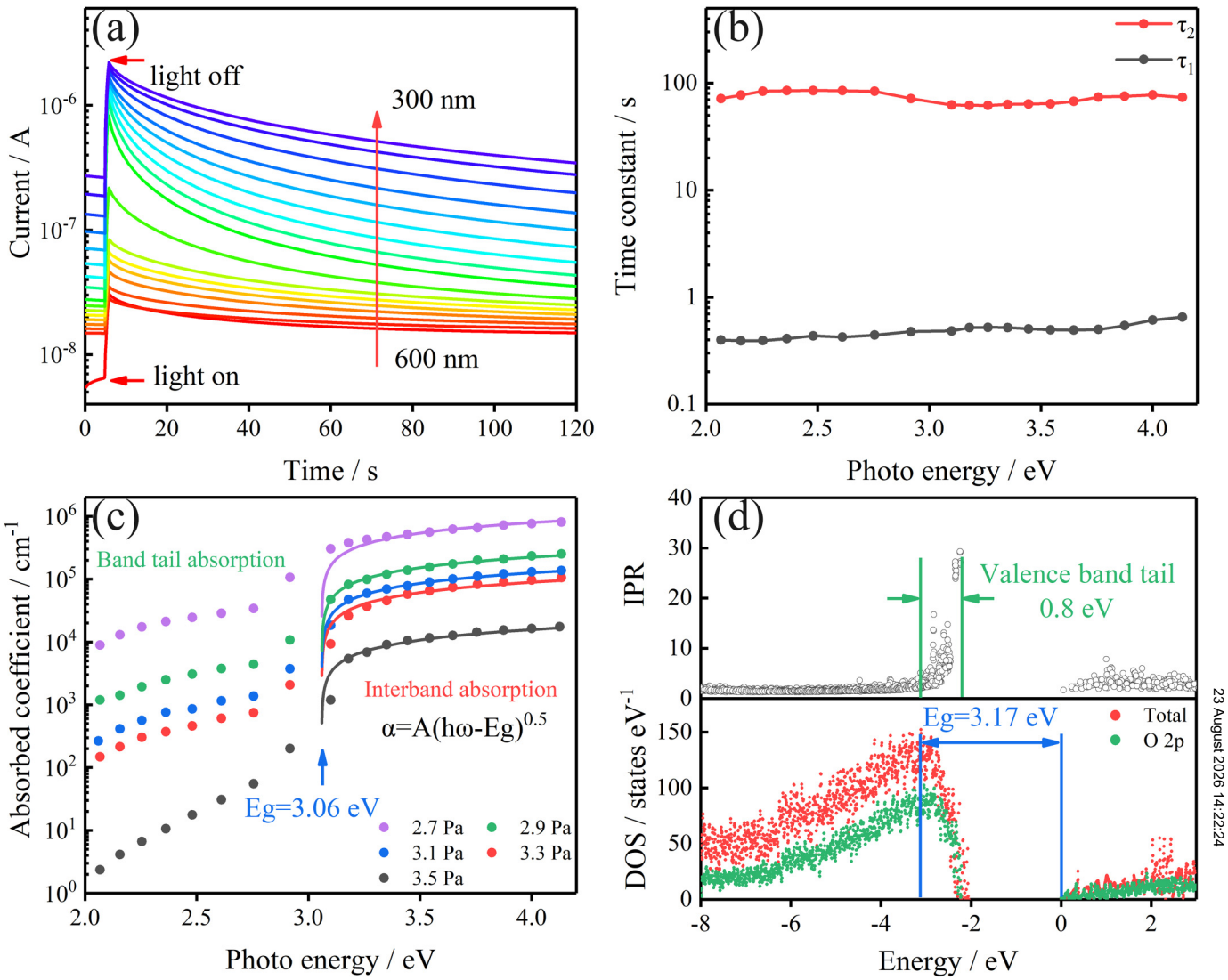



**FIG. 12.** (a) The photocurrent spectrum response of the a-IGZO thin films deposited at 3.1 Pa oxygen pressure, measured in the range of 600–300 nm. (b) The time constants of the decay current with the photon energy extracted from the spectrum response. (c) The absorption coefficient spectra are extracted from the spectrum response of the a-IGZO thin films deposited at 2.7–3.5 Pa oxygen pressure. (d) The inverse participation ratio, total DOS, and the DOS of O2p orbital obtained by DFT calculation.

below the bandgap is down to 2.07 eV. Given that the conduction band tail is small (~150 meV) in a-IGZO[36] and the absorption coefficient increases with the increasing density of oxygen vacancies, this absorption can be attributed to the excitation of a valence band tail from the oxygen vacancies.[23] DFT calculations illustrate that the valance band tail with a width of ~0.8 eV is mainly contributed by the O2p orbital [Fig. 12(d)]. The inverse participation ratio of the band tail is larger than 1, implying the highly localized characteristics.[24] Commonly, the localized valance band tail states stay electrically neutral.[23] When the sample is illuminated, the localized electrons in a valence band tail will be excited onto the conduction band; as a result, tail-like absorption occurs and the neutral oxygen vacancies are ionized.[25,46,47] Moreover, the positive threshold voltage shift of a-IGZO TFTs under positive gate bias stress originates from the electrons trapped in the localized deep acceptor states,[9] which can be suppressed by illumination.[48] Therefore, it suggests that the trapped states sensitive to illumination discussed in this work may be the origin of positive threshold voltage shift.

Figure 13 plots the configuration coordinate diagram. The absorption of continuous light is accompanied by the ionization of the neutral oxygen vacancies ($V_O$) and formation of the excitation

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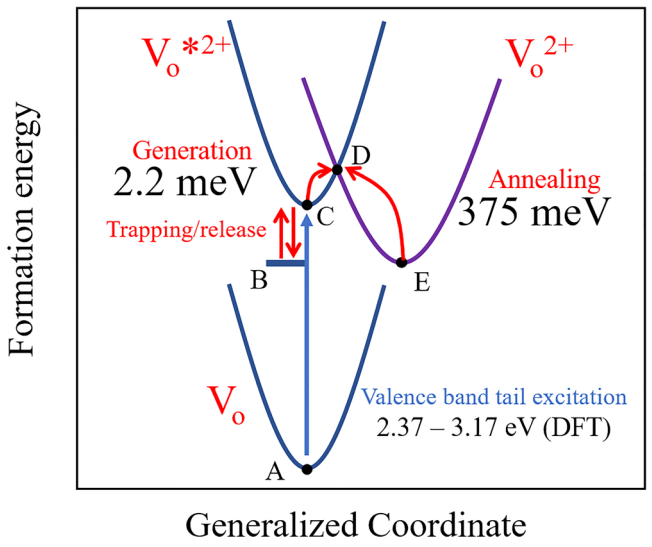



**FIG. 13.** Proposed configuration coordinate diagram for the generation and annihilation of the metastable donors. $V_O$ represents the neutral oxygen vacancies, $V_O^{*2+}$ represents the excited oxygen vacancies, and $V_O^{2+}$ represents the metastable ionized oxygen vacancies. A is the neutral ground state, B is the trap state, C is the excited ionized state, D is the saddle point, and E is the metastable donor state.

states ($V_O^{*2+}$), (A → C). The direct combination of $V_O^{*2+}$ and free carriers (C → A) is delayed by the transition of trapped electrons (C → B and B → C), resulting in short PPC with τ ∼ 0.1 s. It is easier for the excited $V_O^{*2+}$ to transform into the metastable $V_O^{2+}$ with a low energy barrier of 2.2 meV (C → D → E). However, the transformation from $V_O^{2+}$ back to $V_O^{*2+}$ should cross a high energy barrier of 375 meV (E → D → C). This results in long-duration PPC with τ ∼ 100 s. Hereto, the PPC in a-IGZO thin films as well as the instability of a-IGZO TFTs under illumination can be understood by the electron transition from trap states to the conduction band and the metastable donors generated from oxygen vacancies.

## IV. CONCLUSIONS

In this study, a-IGZO thin films are prepared by pulsed laser deposition and fabricated into thin film transistors (TFTs). *In situ* XPS illustrates that the oxygen vacancies in a-IGZO thin films are dependent on the deposition oxygen pressure, and the lower the deposition pressure is, the higher the concentration of oxygen vacancies is. The continuous light (10 s) induced photocurrent includes a fast decay (τ ∼ 0.1 s) and a slow decay (τ ∼ 100 s). However, only the fast decay (τ ∼ 0.1 s) can be observed when illuminated by the pulsed laser (20 ns duration). The fast decay is indeed temperature independent and attributed to the transition of trapped electrons onto conduction band. The numerical simulation and Laplace transformation method indicate that the trap states are at 0.1–0.7 eV below the conduction band. On the contrary, the slow decay is temperature dependent, which is induced by the generation of metastable donors. From the thermally activated current, the donor levels are estimated to be 0.9 and 19.3 meV below the conduction band. The thermal activation energies for the generation and annihilation of the metastable donors are determined to be 2.2 and 375 meV, respectively. The spectrum response measurement and the DFT calculation demonstrate that the metastable donors are ionized from the neutral oxygen vacancies. The PPC and thus the negative shift of threshold voltage of IGZO TFTs under illumination are attributed to the oxygen vacancies.

## SUPPLEMENTARY MATERIAL

See the supplementary material for illustration of PLD configuration, a-IGZO TFT architecture, replicability and reliability of the TFTs performance, amorphous IGZO model and pair-correlations function in MD simulation, the table of atomic percentages, and the table of TFT characteristics.

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## AUTHOR DECLARATIONS

### Conflict of Interest

The authors have no conflicts to disclose.

## DATA AVAILABILITY

The data that support the findings of this study are available within the article.

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